

Module 3

Telecommunication Networks

Syllabus

- **Types of networks:** PSTN, Cellular, Satellite, Internet
 - **Network topologies:** Star, Mesh, Ring, Hybrid
 - Overview of **Switching and Routing**
-

3.1 PSTN — Public Switched Telephone Network

The Public Switched Telephone Network (PSTN) is the oldest, most reliable, and largest global telecommunication network. It was originally designed for real-time voice communication between any two telephone subscribers in the world. It is often referred to as Plain Old Telephone Service (POTS), representing the traditional landline service.

Even though modern networks use digital and packet-based technologies, the concepts of PSTN such as circuit switching, hierarchical exchanges, signaling, and numbering form the foundation of today's telecommunication systems.

Definition

PSTN is the global circuit-switched telephone network that provides analog (and now mostly digital) voice communication by establishing a dedicated connection between subscribers for the entire call duration, using a hierarchical system of exchanges.

Key Characteristics

The PSTN is defined by several core features:

- **Circuit Switching:** It establishes a dedicated physical communication path between callers for the entire call duration.
- **Hierarchical Architecture:** It uses a multi-level system of exchanges (local, tandem, international) to route calls efficiently.

- **Voice-Optimized:** Designed primarily for continuous, real-time analog voice signals, ensuring predictable voice quality with low delay and low jitter.
- **Standardized:** It Follows global standards for numbering (E.164) and signaling (SS7). Uses out-of-band SS7 signaling for call setup, routing, and teardown.
- **Backbone** is fully digital (PCM/TDM, SDH/SONET, optical fiber) while the local loop may still be analog copper.
- **High Reliability:** It is an extremely mature and stable system, built for consistent voice quality and availability.

Block Diagram of PSTN

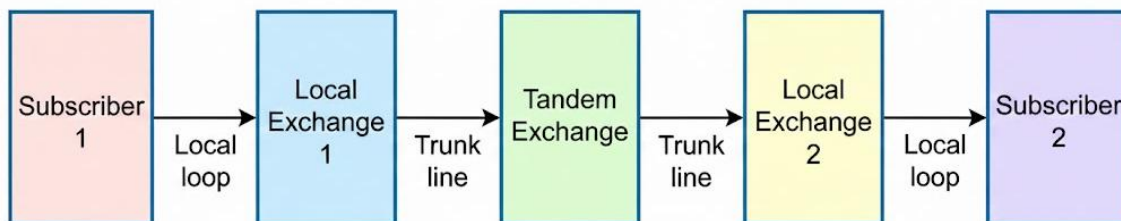


Fig. Block diagram of PSTN

Key Components and Architecture

The PSTN is built from specific parts arranged in a hierarchy:

- **Subscriber:** The end user equipment (handset); typically analog, converting sound to electrical signals.
- **Local Loop (Subscriber Line):** The twisted pair copper wire connecting the subscriber premises to the nearest Local Exchange.
- **Local Exchange / Central Office (CO):** The lowest level of switching hierarchy. It terminates the subscriber lines. It provides dial tone, detects dialed digits, and establishes local calls. It routes long-distance calls to higher-level switches.
- **Tandem/Transit Exchange:** This is intermediate switching centers used for interconnecting multiple local exchanges to route calls between local exchanges that do not have direct links between them, especially for long-distance calls.

- **Trunks (Inter-Exchange Links):** These are high-capacity transmission links (like E1/T1 digital lines or fiber optics) that carry multiple voice channels using TDM (Time Division Multiplexing).

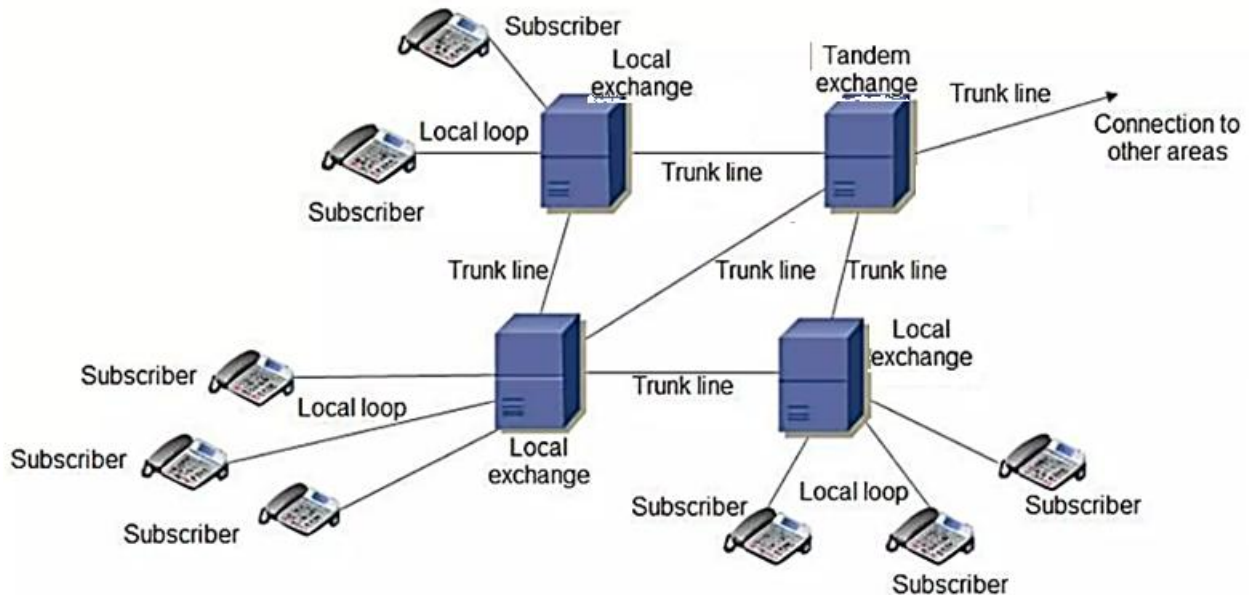


Fig. PSTN: Key components and architecture

- **Switching Equipment:** Digital circuit switches that establish and release circuits for each call.
- **Signaling System (SS7):** This is a separate, intelligent network that handles all the control tasks—call setup, routing, billing, and teardown—without using the voice path.
- **Numbering and Routing:** Uses E.164 numbering plan (Country Code + National Number). Exchanges analyze digits to determine routing.

Working Principle: Circuit Switching

The operation is based on establishing a dedicated circuit:

1. Call Setup:

- **Dialing:** The caller dials the number, and the digits are received by the Local Exchange (CO).
- **Digit Analysis & Routing:** The CO uses the destination number (E.164) to determine the next switch (CO or Tandem) to which the call must be routed.

- **Signaling (SS7):** The exchanges communicate using SS7 messages to exchange information, reserve a path, and confirm the route.
- **Circuit Establishment:** A physical or logical channel (e.g., a specific time slot in a TDM frame) is allocated and reserved end-to-end.
- **Connection:** Once the dedicated circuit is established, the ringing signal is sent to the called party.

2. Call in Progress

- The circuit remains reserved exclusively for the duration of the call.
- Ensures constant bandwidth (64 kbps PCM) and consistent voice quality.

3. Call Termination / Call Teardown:

- When either the caller or receiver hang up, SS7 signals are transmitted to all intermediate exchanges, commanding them to release the dedicated circuit and free the trunks/time slots for subsequent calls.

Advantages

- **High Predictable Quality:** Dedicated circuit ensures clear voice with no delay variation (no jitter).
- **High Reliability:** Very robust infrastructure with backup power.
- **Simple Model:** Easy to understand, use, and bill.
- **Works Without Local Power:** Phones are powered from the exchange.

Disadvantages

- **Inefficient Bandwidth Use:** The dedicated circuit is wasted during periods of silence in a conversation. It is poor for bursty data.
- **Slow Setup:** Establishing the dedicated path takes time (several seconds for long-distance).
- **Expensive to Scale:** Building and maintaining dedicated circuits for every call is costly.
- **Limited Services:** Primarily for voice. Not natively suited for modern high-speed data or video.

Applications

- Traditional landline voice calls (POTS).
- Fax machine communication.
- Dial-up internet access (historical).

- Backup communication link for alarms and emergency systems.
- As an interconnection backbone for other networks (mobile, VoIP) via gateways.

3.2 Cellular Network

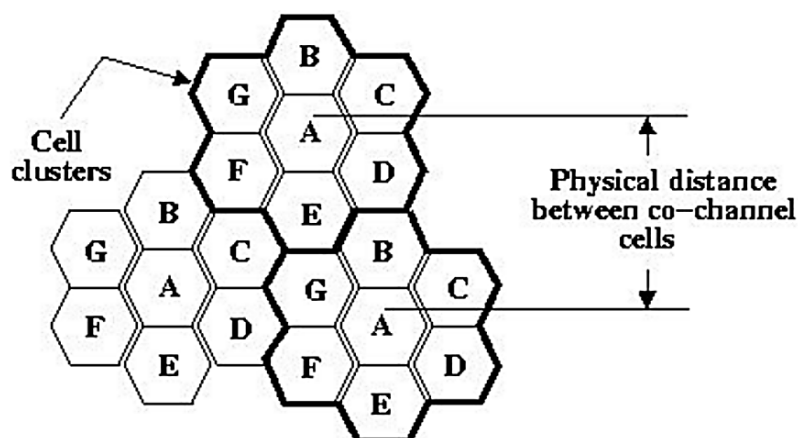
The Cellular Network, commonly known as the mobile phone network, is a wireless telecommunication system designed to provide voice and data services to users with mobility. It represents the most significant evolution from fixed-line (PSTN) telephony, enabling communication from virtually anywhere.

The core innovation is the division of a large area into small, interconnected zones called "cells", each served by its own low-power Base Station. This structure allows frequency reuse, enabling millions of users to communicate simultaneously using limited radio spectrum.

Cellular systems form the foundation of today's mobile communication technologies—2G, 3G, 4G LTE, and 5G.

Definition

A Cellular Network is a wireless communication network where a geographic region is divided into small hexagonal or irregular-shaped areas called cells. Each cell is served by at least one fixed-location transceiver known as a Base Station (BS) or Cell Site. These cells collectively provide radio coverage over a wide area, enabling mobile devices to communicate while moving through different cells without losing the connection.



Key Characteristics

Cellular networks are defined by the following features:

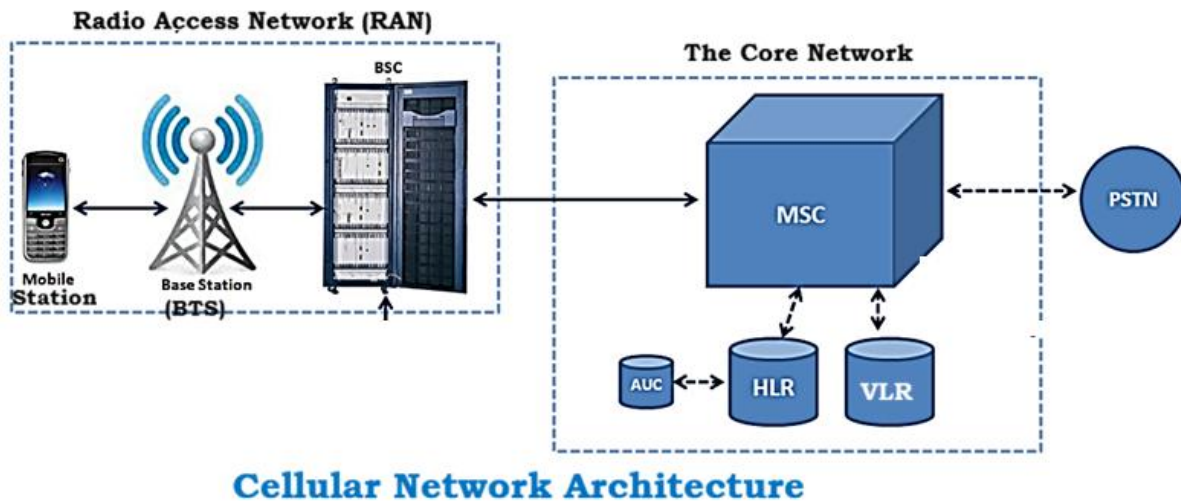
- **Cellular Architecture:** The entire network coverage area is divided into small cells (typically 1–20 km radius), each with its own base station.
- **Frequency Reuse:** The same set of radio frequencies is reused in multiple, non-adjacent cells, greatly improving spectrum efficiency. This results in a significant increase in the total number of simultaneous users that can be supported by the system.
- **Mobility Support:** The network automatically transfers an ongoing call or data session from one cell to another as the user moves through a process called Handoff (or Handover). Thus, the users can move freely while staying connected without dropping the call.
- **Hierarchical Network Structure:** Comprises the Radio Access Network (RAN), forming the cellular coverage, and the Core Network for switching, mobility management, and interconnection with other networks. Components like BTS, BSC, and MSC work together to provide efficient call control and mobility management.
- **Digital Signaling and Switching:** Modern cellular networks (from 2G onwards) use digital modulation, digital switching, and packet-based technologies, encryption, and advanced multiplexing for efficient and secure use of spectrum.
- **Supports Voice and Data:** Provides services beyond voice, including SMS, mobile internet, multimedia, and location-based services.
- **Standardization:** Cellular systems follow internationally standardized protocols defined for different generations (1G, 2G, 3G, 4G LTE, 5G).

Key Components and Architecture

The cellular network consists of multiple subsystems:

1. Mobile Station (MS)

- The user equipment (e.g., mobile phone, smartphone, data card).
- It contains a transceiver, antenna, and a Subscriber Identity Module (SIM) for user identification and authentication.
- It performs transmission and reception.
- It measures the signal strength from surrounding base stations (BSs) and reports to the network.



2. Base Transceiver Station (BTS)

- Fixed radio transceiver and antenna located in each cell.
- Handles radio transmission and reception between mobile station (MS) and network.
- Provides coverage and assigns radio channels.
- Handles immediate cell traffic.

3. Base Station Controller (BSC)

- Controls multiple base transceiver station BTSs.
- Manages radio resources (frequency allocation, power control).
- Coordinates the Handoff process between adjacent cells it controls.

4. Mobile Switching Center (MSC)

- Central switching unit for cellular networks.
- It also connects to the PSTN and other networks.
- Performs call routing, mobility management, handover control, and billing coordination.

5. Home Location Register (HLR)

- Database storing permanent subscriber information (phone number, plan, authentication keys).

6. Visitor Location Register (VLR)

- A temporary database linked to the MSC that stores information about visiting subscribers currently roaming within its service area.

7. Authentication Center (AuC)

- Provides authentication and encryption keys to secure communication.

8. Gateway MSC (GMSC)

- Connects cellular network to external networks (PSTN, ISDN, VoIP).

Working Principle of Cellular Networks

The operation of a cellular network is governed by two fundamental principles – Frequency Reuse (for capacity) and Handoff (for mobility) – and is executed through a step-by-step call flow managed by the switching hierarchy.

Frequency Reuse (Capacity Management):

The coverage area is divided into a cluster of cells (e.g., 7 cells) that use the entire set of allocated frequencies. Due to the use of low-power transmitters in small cells, this same set of frequencies is reused in identical clusters located sufficiently far away. This strategic repetition multiplies the network's capacity, allowing it to serve millions of users with limited spectrum. The physical separation distance of the cluster of cells must be large enough to limit Co-Channel Interference (CCI).

Handover (Handoff) (Mobility Management):

Handover is the seamless process of automatically transferring an ongoing call or data session from the current Base Station to a neighboring Base Station when the mobile device moves across cell boundaries. As the user moves towards the edge of a cell, the signal strength from the serving BTS decreases. The Mobile Station (MS) constantly measures the signal strength of its serving cell and surrounding cells, reporting these measurements to the Base Station Controller (BSC) and Mobile Switching Center (MSC). When the signal from the current cell falls below a threshold, and the signal from an adjacent cell is deemed strong enough, the MSC initiates the handover to ensure continuity without dropping the call.

Types:

- **Hard Handoff (Break-Before-Make):** The connection to the old cell is terminated before the connection to the new cell is established (used primarily in 2G GSM and 4G LTE).
- **Soft Handoff (Make-Before-Break):** The mobile is connected to two or more Base Stations simultaneously during the brief transition, guaranteeing a seamless, interruption-free transfer (used typically in 3G CDMA networks).

Call Flow Mechanics (Step-by-Step)

The principles of cellular communication are implemented through the following detailed call procedure, involving precise signaling and resource allocation:

1. Initial Connection and Location Management

- **Initial Registration:** When powered on, the Mobile Station (MS) scans for an appropriate Broadcast Control Channel (BCCH), synchronizes with the selected BTS, and sends a registration (location update) request. BCCH (Broadcast Control Channel) is the channel through which a Base Transceiver Station (BTS) continuously broadcasts important system information to all mobile stations (MS) within the cell. It allows a mobile to identify the cell, synchronize with it, and understand how to access the network.
- **Location Update:** If the MS moves into a new Location Area (group of cells), it periodically updates its location. The MSC updates the temporary location in the Visitor Location Register (VLR) and informs the central Home Location Register (HLR). This ensures the network knows where to find the user.

2. Call Setup

- **Dialing:** The user dials the number, and the call request (containing the dialed digits) is sent from the mobile station (MS) to the nearest Base Transceiver Station (BTS).
- **Forwarding to Core Network:** BTS forwards this request to the Base Station Controller (BSC) which controls multiple BTSs. The BSC forwards it to the Mobile Switching Centre (MSC). BTS → BSC → MSC.
- **Authentication and Allocation:** The MSC checks the subscriber's identity and service authorization using the HLR/VLR database. HLR (Home Location Register) – permanent subscriber details and VLR (Visitor Location Register) – temporary information for roaming users.

- **Destination Analysis:** The Mobile Switching Centre (MSC) analyses the dialed number and determines whether the call should be routed to: (i) another mobile within the same network, (ii) another mobile network, or (iii) a PSTN landline.

3. Radio Channel Allocation

Once the call is authenticated, the BSC assigns an available frequency channel or time slot to the user and a dedicated radio link is created between the Mobile Station (MS) and the BTS, enabling the call to begin.

4. Call in Progress and Continuity

- **Communication:** Voice or data traffic flows continuously over the assigned radio channels.
- **Quality Management:** The network performs continuous power control (adjusting the MS and BTS transmission power) and interference management to maintain call quality.
- **Mobility:** When the serving cell's signal becomes weak, the BSC (or MSC) triggers a handoff to a neighboring BTS without dropping the call.

5. Call Termination

When either user ends the call:

- MS sends a call release message to BTS.
- BTS → BSC → MSC release switching resources.
- The MSC sends a clear command to the BSC to immediately release the radio resources (frequency/time slot) and core switching resources.
- The BSC then releases the assigned radio channel and makes it available for new users.

3.3 Satellite Networks

Satellite networks are wireless communication systems that use artificial satellites orbiting the Earth to provide long-distance communication services. The satellite networks can cover very large geographical areas, including oceans, deserts, and remote regions where wired or cellular infrastructure is impractical. Satellite communication plays a crucial role in global connectivity by enabling broadcasting, navigation, emergency communication, and international telephony and data services.

Definition

A Satellite Network is a wireless telecommunication system utilizing one or more satellites, placed in Earth's orbit, as active relay stations (transponders). These satellites receive, amplify, and retransmit signals between ground-based transmitters and receivers, known as Earth Stations, enabling reliable, long-distance communication across extensive geographical coverage areas (footprints).

Necessity

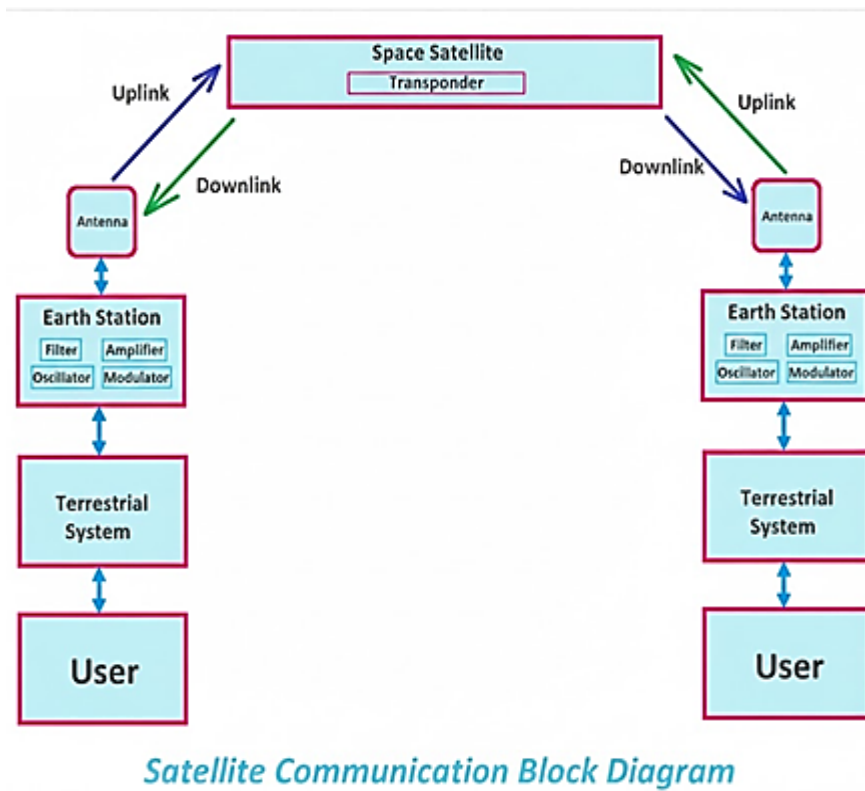
They are essential for applications requiring coverage over extremely wide geographical areas (e.g., oceans) where laying fiber optic cables or setting up terrestrial radio systems is impractical or impossible.

Key Characteristics / Features:

Satellite networks are defined by several distinct features:

- **Extensive Coverage (Footprint):** A single satellite in high orbit can cover up to one-third of the Earth's surface, providing connectivity across continents and oceans.
- **High Initial Cost, Low Cost-Per-User:** Launching and building satellites is extremely expensive, but once operational, the cost of serving a large number of users over a wide area becomes relatively low.
- **Broadcast Capability:** Inherently supports point-to-multipoint (broadcast) communication, making it ideal for distributing TV and radio signals to millions of receivers simultaneously.
- **Propagation Delay (Latency):** Signals must travel to space and back, causing a significant delay. For Geostationary Earth Orbit (GEO) satellites, this round-trip delay is about 0.5 seconds, which is noticeable in voice and data conversations.
- **Transparent Operation:** The satellite typically acts as a "bent pipe" or repeater in the sky. It receives signals from Earth, shifts their frequency to avoid interference, amplifies them, and beams them back down without complex processing.
- **Vulnerability to Environmental Effects:** Signal quality can be degraded by atmospheric conditions, especially at higher frequencies (rain fade, solar interference).

Block Diagram of Satellite Communication



Key Components of a Satellite Network

A basic satellite communication link consists of three main segments:

1. Space Segment (Satellite)

The orbiting satellite is the central element of the network. Its key parts are:

- **Transponders:** Each transponder receives an uplink signal, amplify it, and perform a frequency translation to prevent receive-transmit interference (desensitization) before retransmitting the downlink signal.
- **Antenna System:** Responsible for receiving the weak uplink signal from Earth and transmitting the processed amplified downlink signal back to earth.
- **Power System:** Consists of Solar Arrays and batteries to generate and store electrical power necessary for all on-board equipment.
- **On-board Control System:** Maintains satellite position, orientation, and operation.

2. Ground Segment (Earth Stations)

This segment comprises all ground-based transmitting and receiving equipment:

- **Gateway/Hub Stations:** Large, fixed Earth stations with high-power transmitters and large parabolic dish antennas (typically 5m to 18m diameter). They interface the satellite network with terrestrial networks (like PSTN or the Internet backbone).
- **Very Small Aperture Terminal (VSAT):** A small, user-owned dish antenna (typically 0.75m to 2.4m in diameter) used for receiving or two-way communication (e.g., for bank ATMs, rural internet).

3. Control Segment

It is used to monitor the satellite's health, track its position, and send commands for orbital correction or configuration changes.

Types of Satellite Orbits

The operational characteristics of a satellite network, particularly latency and coverage, are determined by the satellite's orbit altitude.

1. GEO – Geostationary Earth Orbit

- Altitude: ~35,786 km above Earth.
- Appears stationary relative to Earth.
- Coverage: Provides maximum coverage area (one satellite covers approximately 1/3rd of Earth's surface).

Applications:

TV broadcasting, weather monitoring, long-distance telephony.

Limitation:

High latency (propagation delay ~250 ms one-way).

2. MEO – Medium Earth Orbit

- Altitude: 2,000–20,000 km.
- Moderate delay and coverage.

Applications:

Navigation systems (GPS, Galileo).

3. LEO – Low Earth Orbit

- Altitude: 500–2,000 km.
- Low latency (<50 ms) and high signal quality.
- Requires large satellite constellations.

Applications:

Mobile satellite services, broadband internet (Starlink, OneWeb).

Working Principle of Satellite Networks

The fundamental operation involves a three-step relay process:

1. Uplink Transmission (Earth to Space):

The transmitting Earth Station (e.g., a Gateway or VSAT) converts the baseband information (voice, video, data) into an appropriate radio frequency (RF) signal. This signal is then transmitted to the satellite using a high-power antenna on a specific Uplink Frequency. Uplink frequency is higher than downlink frequency to avoid interference.

2. Satellite Transponder Processing (In Space):

The satellite's antenna receives this weak uplink signal. A transponder selects the signal, amplifies it greatly, and changes the signal's carrier frequency from the uplink frequency to a lower downlink frequency. This prevents the powerful downlink signal from interfacing with the weak incoming signal.

3. Downlink Transmission (Space to Earth):

The amplified and frequency shifted signal is transmitted back to one or more Earth stations. Downlink covers a defined geographical area called the footprint. All receiving Earth stations or VSATs within this footprint can pick up the signal. The receiving Earth station captures the signal, amplifies it and demodulates it to recover the original information (voice, video, or data).

Advantages

- **Wide Area Coverage:** Can serve remote, inaccessible, rural, and maritime regions unreachable by terrestrial networks (e.g., deserts, islands, ships).
- **Reliability:** Once in orbit, the system is not susceptible to ground-based disasters like earthquakes or floods along the path. Works even when ground infrastructure is damaged.

- **High bandwidth capacity:** High bandwidth capacity, ideal for simultaneous broadcasting (point-to-multipoint communication).
- **Broadcast/Multicast Efficiency:** Ideal for distributing the same content (TV, radio, software updates) to a vast number of locations.
- **Uniform Performance:** Quality of service is consistent across the entire coverage area, independent of distance.
- **Rapid Deployment:** Quick deployment for emergency services or temporary network extensions (e.g., disaster relief). No need for physical cabling.

Disadvantages

- **High Initial Cost:** Extremely high initial cost for manufacturing, launching, and maintaining satellites and control stations.
- **High Propagation Delay (in GEO):** The ~0.5 second latency makes two-way voice/data conversation less natural and affects real-time interactive applications.
- **Susceptibility to Signal Fading:** Atmospheric conditions (rain, snow, clouds) can attenuate signals.
- **Complex Coordination and Licensing:** Requires careful international coordination to avoid frequency interference and orbital slot conflicts.

Applications

- Television and Radio Broadcasting (DTH - Direct-to-Home).
- Long-Distance Telephony (especially for connecting remote areas to the global PSTN).
- Satellite Internet Access for remote enterprises, ships (maritime VSAT), and homes.
- Navigation Systems: Forming the core of Global Positioning Systems (GPS) and other Global Navigation Satellite Systems (GNSS).
- Weather Forecasting and Earth Observation using remote sensing satellites.
- Military and Strategic Communication.

3.4 Internet Networks

The Internet is not a single, unified network; rather, it is a global, public, interconnected network of networks. It is the world's largest and most widely used communication system. Unlike the Public Switched Telephone Network (PSTN) and other traditional circuit-switched systems, the Internet is a packet-switched, data-centric network designed to efficiently transmit data among millions of interconnected devices worldwide.

The Internet is a general-purpose packet-switched data network capable of carrying all forms of digital information—including voice, video, text, and data files. It enables a wide range of services such as the World Wide Web (WWW), email, file transfer, video streaming, Voice over Internet Protocol (VoIP), and cloud computing. As such, the Internet forms the backbone of modern digital communication and information exchange.

Definition

The Internet is a worldwide, decentralized, packet-switched network that uses standardized communication protocols—primarily the Internet Protocol Suite (TCP/IP)—to interconnect billions of devices across diverse physical networks, including fiber-optic backbones, cellular networks, and satellite links. It provides a universal platform for reliable data communication and the delivery of digital services.

Basic Concept of Internet Communication

The Internet's architecture and operation are based on three core technical concepts:

Packet Switching

Packet switching is the fundamental principle of data transmission over the Internet. In packet switching, information (data, voice, or video) is segmented into small, independent blocks called packets. Each packet contains a small portion of the message data along with control information, including the source and destination addresses. These packets travel independently through the network, possibly taking different routes, and are reassembled into the original message at the destination. This method allows many users to share network capacity efficiently, as a dedicated link is not required for the entire communication duration (unlike circuit switching, e.g., PSTN).

TCP/IP Protocol Suite

TCP/IP (Transmission Control Protocol/Internet Protocol) is the universal, standardized language of the Internet. This suite governs how data is broken

down, addressed, transmitted, routed, and received across the entire global network. IP (Internet Protocol) governs addressing (logical addressing using IP addresses) and the routing of packets from the source network to the destination network. TCP (Transmission Control Protocol) ensures reliable data delivery. It manages the reassembly of packets at the destination, detects and corrects errors, and ensures data is received in the correct order without loss.

Routers

Routers are intelligent networking devices that form the backbone of the Internet. Routers examine the destination address of every incoming packet. They dynamically use complex routing tables and algorithms to forward the packet along the most optimal, least-congested path across the global network toward its final destination. They connect different networks (e.g., a home network to an ISP, or one ISP's backbone to another) to build the Internet's structure.

Key Characteristics

The Internet is defined by its foundational principles:

- **Packet Switching:** Data is broken into small, addressed packets that travel independently across the network, taking potentially different routes to the destination where they are reassembled.
- **Decentralized Architecture:** There is no central controlling authority. It is a mesh of Autonomous Systems (networks operated by ISPs, companies, universities) that cooperate via standardized protocols.
- **Best-Effort Delivery:** The core IP protocol does not guarantee delivery, order, or timing of packets. Reliability is managed by higher-layer protocols like TCP.
- **Scalability and Robustness:** Its distributed design allows it to grow organically and automatically route around failures.
- **Protocol-Based Interoperability:** Universal adherence to open standards (TCP/IP, HTTP, SMTP) allows any device, on any underlying network, to communicate.
- **Convergence Network:** It carries all forms of media (data, voice as VoIP, video as streaming), leading to the convergence of services.

Key Components and Architecture

- **End Systems (Hosts):** The source and destination devices – such as computers, smartphones, servers – that generate and receive data. End

systems run network applications, including web browsers, email clients, and multimedia applications.

- **Communication Links:** The physical media – such as fiber optics, coaxial cable, radio links – that provide connectivity between end systems and the Internet.
- **Routers:** The most critical hardware component. Routers are specialized computers that interconnect different networks. They examine packet headers and route them towards their final destination by choosing the best path through the interconnected network of networks.
- **Internet Service Providers (ISPs):** Organizations that provide access to the Internet. They are arranged in a hierarchy: •
 - **Tier-1 ISPs:** Global backbone providers with extensive international connectivity
 - **Tier-2 ISPs:** Regional or national providers that connect to Tier-1 networks
 - **Tier-3 / Local ISPs:** Retail providers that deliver Internet access directly to end users
- **Protocols:** The rules governing communication. The core suite is TCP/IP.
 - **IP (Internet Protocol):** Responsible for addressing and routing packets across networks (the "postal system").
 - **TCP (Transmission Control Protocol):** Provides reliable, ordered, error-checked delivery of a stream of data between applications (ensures "letters" arrive and are in order).
 - **Other Protocols:** HTTP (web), SMTP (email), DNS (translates names to IP addresses).

Working Principle: Packet Switching and Routing

Data Segmentation:

When a user sends data (such as a file or email), the data is divided into smaller units called packets. Each packet is assigned a header containing the source and destination IP addresses, along with sequence numbers and other control information.

Routing Decision:

Each packet is launched into the network. Every router that receives a packet examines the destination IP address in the packet header. Using

its routing table (which represents known network paths), the router determines the next hop—the best immediate link to forward the packet closer to its final destination. This routing decision is made independently for each packet.

Independent Travel:

Packets belonging to the same message may traverse different paths through the network depending on current network traffic, network congestion, link availability and routing protocol decisions

Reassembly:

When all packets reach the destination host, they are reassembled in the correct order using the sequence numbers. If packets are lost or corrupted, the Transmission Control Protocol (TCP) detects the errors and requests retransmission. Once all packets are successfully received, the original data is reconstructed.

Advantages of Internet Networks

- ***Efficient Bandwidth Utilization:*** Uses packet switching, allowing multiple users to share network resources dynamically, which avoids wasting bandwidth during idle periods.
- ***Supports Multiple Services:*** Enables diverse services such as data transfer, voice communication, video streaming, cloud services, and IoT on a single unified network.
- ***Highly Scalable:*** The Internet can easily expand to support billions of devices and users without requiring major architectural changes.
- ***Cost-Effective:*** Shared infrastructure and statistical multiplexing make Internet communication cheaper compared to dedicated circuit-switched networks.
- ***Flexible and Adaptive Routing:*** Dynamic routing allows packets to take alternate paths during congestion or link failure, improving network robustness.

Disadvantages of Internet Networks

- ***No Guaranteed Quality of Service (QoS):*** Delay, jitter, and packet loss may occur because the Internet follows a best-effort delivery model.
- ***Security Vulnerabilities:*** Open connectivity exposes the network to threats such as hacking, malware, phishing, and denial-of-service attacks.

- **Congestion and Performance Variations:** Network performance depends on traffic load, leading to reduced speed during peak usage times.
- **Complex Network Management:** Efficient operation requires sophisticated protocols for routing, congestion control, error handling, and security.
- **Dependence on Power and Infrastructure:** Internet services rely heavily on continuous power supply and physical infrastructure, making them vulnerable during failures or disasters.

Applications of Internet Networks

- **Web Browsing and Email:** Provides global access to information and supports instant electronic communication across the world.
- **Cloud Computing and Storage:** Enables on-demand access to remote computing resources, data storage, and software services.
- **Video Streaming and Online Gaming:** Supports real-time and on-demand multimedia services requiring high data rates and low latency.
- **Voice over IP (VoIP):** Allows voice communication over packet-switched networks, reducing cost compared to traditional telephony.
- **Social Networking and Collaboration:** Facilitates social interaction, content sharing, and real-time collaboration across geographical boundaries.
- **Internet of Things (IoT):** Connects smart devices and sensors, enabling automation, monitoring, and intelligent decision-making systems.

3.5 Network Topologies

Network topology refers to the physical or logical arrangement of devices (nodes) and the communication links between them in a network.

It defines how computers, routers, switches, and other devices are interconnected to form a functional communication system and how data flows from one device to another.

Choosing the right topology is a fundamental design decision, as it directly impacts the network's cost, performance, reliability, and scalability. Topologies can be physical (the actual layout of cables and devices) or logical (the path data takes through the physical layout).

Physical vs. Logical Topology

It is important to distinguish between these two aspects:

- **Physical Topology:** This describes the actual physical layout of the cables, the placement of nodes, and the path of the wiring. (e.g., how the wires are physically connected into a central switch).
- **Logical Topology:** This describes the path that the data signals follow as they move from one node to another, regardless of the physical layout. (e.g., a network might look like a star physically but act like a ring logically).

Types of Network Topologies

Broadly, topologies are classified based on the "shape" of the connections. The major types are:

- **Star Topology:** Nodes are connected to a central hub.
- **Mesh Topology:** Nodes are connected to multiple other nodes for high redundancy.
- **Ring Topology:** Nodes are connected in a closed circular loop.
- **Hybrid Topology:** A combination of two or more different topologies.

Each topology has distinct characteristics, advantages, and disadvantages, making it suitable for specific applications.

3.6 Star Topology

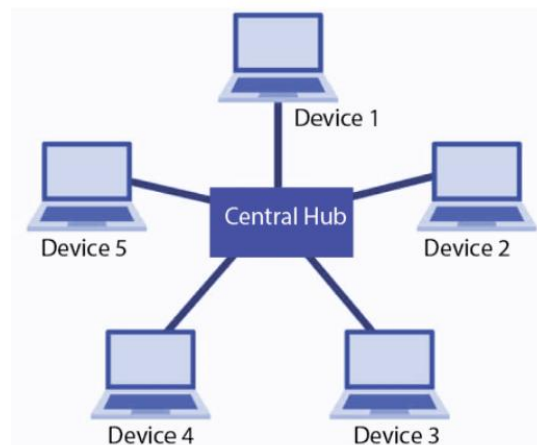
Star topology is a network configuration in which all devices (nodes) are connected individually to a single central device (such as a hub, switch, router, or server), creating a layout that resembles a wheel with spokes.

All communication between devices must pass through this central point. Peripheral devices do not connect directly to each other. It is the most widely used topology in modern Local Area Networks (LANs), especially Ethernet-based networks in homes, offices, and institutions.

Core Structure and Design

- **Centralized Control:** A single central device—typically a switch, hub, or router—manages all data traffic and communication.

- **Point-to-Point Links:** Each individual node (computer, phone, or peripheral) has a dedicated, independent connection to the central controller / device.
- **No Direct Peer Communication:** Nodes do not connect to one another. Communication between Node A and Node B must always pass through the central controller.



Star Topology

Key Characteristics

- **Centralized Control:** The central device manages data transmission, monitoring, and control.
- **Point-to-Point Connections:** Each node has its own dedicated cable or wireless link to the central device.
- **Dependence on Central Device:** The entire network relies on the central hub/switch for operation.
- **Easy Installation and Expansion:** Nodes can be added or removed without disrupting the rest of the network.
- **Fault Detection and Isolation:** Failure in one cable or node affects only that node and is easy to identify.

Data Flow (How It Works)

1. **Initiation:** A source node transmits data intended for a destination node.
2. **Central Processing:** The data reaches the central device (the "intelligent traffic controller").

3. **Relay mechanism:**

- If using a **Switch**: The device examines the destination address and forwards the data **only** to the specific port of the intended recipient.
- If using a **Hub**: The device broadcasts the data to all connected nodes (less efficient).

4. **Delivery**: The destination node receives the data via its dedicated link.

Advantages of Star Topology

- **Ease of Scalability**: Adding or removing devices is plug-and-play. You simply connect a new cable to an open port on the central device without disrupting existing users.
- **Fault Isolation and Reliability**: Failure of one node or cable does not affect other nodes.
- **Simplified Troubleshooting**: Dedicated links make it easy to locate and fix faults.
- **High Performance (with Switches)**: Switch-based star networks eliminate collisions and allow simultaneous data transfers.
- **Centralized Network Management**: Security, monitoring, and administration are easier from a single point.

Disadvantages of Star Topology

- **Single Point of Failure**: Failure of the central hub/switch causes the entire network to go down.
- **Higher Cabling Cost**: Requires more cables than bus or ring topologies because each node needs a separate link.
- **Limited Scalability**: Expansion depends on the number of available ports and processing capacity of the central device.
- **Performance Bottleneck**: If the central device is overloaded or is a simple hub, performance degrades as traffic increases.

Applications of Star Topology

- **Computer Networks**: Office LANs, College and school computer laboratories, home networks using Ethernet switches or Wi-Fi routers
- **Telecommunications**: Public Switched Telephone Network (PSTN)- Individual telephones connect directly to a local exchange, PBX Systems- Telephone extensions connect to a central private exchange.

- **Wireless Networks:** Wi-Fi networks where a Wireless Access Point (WAP) acts as the central hub for multiple devices.

3.7 Mesh Topology

Mesh topology is one of the most reliable and robust network topologies. In this design, devices are interconnected through multiple dedicated point-to-point links, creating several alternative paths for data transmission. This high level of redundancy provides exceptional fault tolerance, making it ideal for critical communication systems—such as internet backbones, military networks, and telecommunication cores—where uninterrupted connectivity is essential.

Definition

Mesh Topology is a network configuration in which each node (device) is connected directly to many other nodes, either fully or partially, through dedicated point-to-point links. This interconnection provides multiple, redundant paths for data to travel from any source to any destination, ensuring high reliability, fault tolerance, security and continuous operation even if individual links or nodes fail.

Types of Mesh Topology

Mesh topology is classified based on the degree of interconnection between nodes.

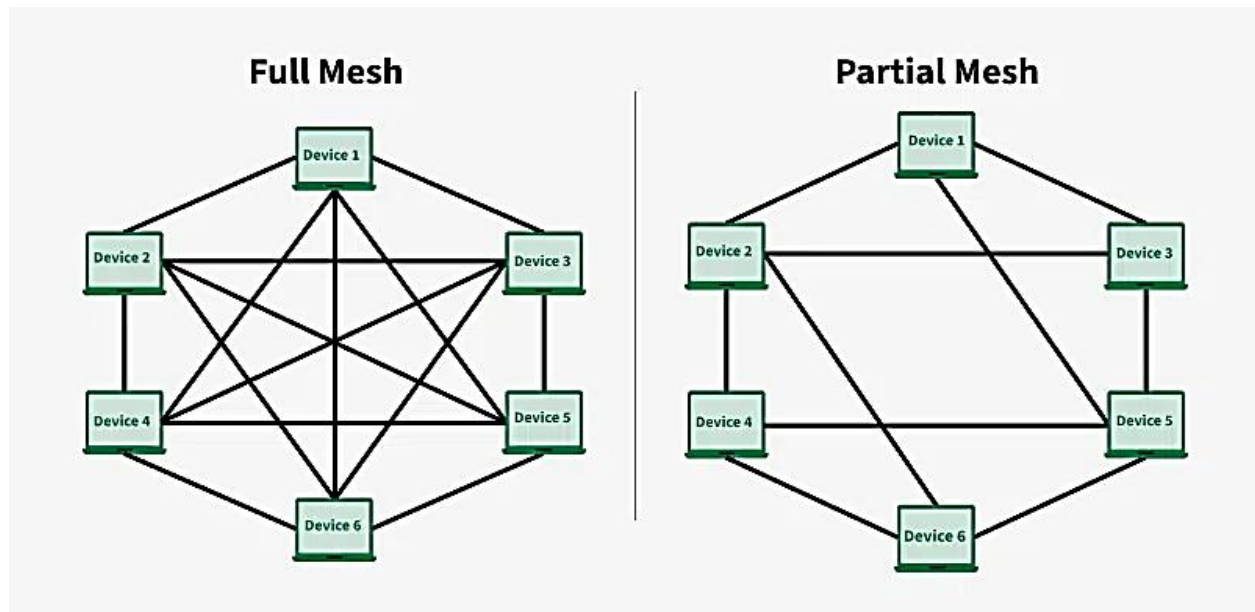
a) Full Mesh Topology

- Every node is directly connected to all other nodes in the network.
- Provides maximum redundancy and reliability.
- Requires a very large number of links and I/O ports.
- Best suited for core backbone networks where downtime is unacceptable (e.g., critical financial or military hubs).

b) Partial Mesh Topology

- Only important or critical nodes are fully interconnected.
- Remaining nodes connect only to those devices with which they frequently exchange data.
- Reduces cost and cabling complexity while maintaining reliability for essential paths.

- Best suited for large networks requiring high reliability with controlled cost. Commonly used in Wide Area Networks (WANs) and Internet Service Provider (ISP) architectures.



Structure and Key Characteristics

- **Decentralized Structure:** There is no central controlling device (like a hub or switch). Nodes communicate directly or via other nodes.
- **Point-to-Point Links:** Each connection is a dedicated link between two nodes.
- **Node Intelligence:** Each node must possess routing logic to participate in forwarding data for the network.
- **Mathematical Scalability Limitation (for Full Mesh):** The number of physical links grows rapidly with the number of nodes.
 - Formula for Total Links in a Full Mesh: $L = n(n-1)/2$ (where n is the number of nodes).
 - Ports per Node: $n-1$ ports are required on each device.
 - This exponential growth makes full mesh impractical for large networks.

Working of Mesh Topology

Mesh networks operate on principles of dynamic, decentralized routing:

1. **Autonomous Routing:** Each node acts as an independent router. It contains a routing table and logic to determine the best path for data.

2. **Dynamic Path Selection:** When a node sends data, it can choose the most direct or the least congested route to the destination from among the available multiple paths.
3. **Self-Healing (Fault Tolerance):** If a primary link fails, the routing protocols in the nodes automatically detect the failure and recalculate paths in real-time, rerouting data through an alternative link without interrupting the communication session. This is the core of its robustness.

Advantages of Mesh Topology

- **High Reliability and Fault Tolerance:** Multiple redundant paths ensure communication even during failures.
- **No Single Point of Failure:** Failure of one link or node does not affect the entire network.
- **High Security:** Dedicated links make it difficult for unauthorized nodes to intercept traffic.
- **No Traffic Congestion:** Each connection has independent bandwidth.
- **Robust Network Performance:** Stable and uninterrupted data transmission.
- **Easy Fault Isolation:** Faulty links can be identified without shutting down the network.

Disadvantages of Mesh Topology

- **Very High Cost:** Requires extensive cabling and multiple network interfaces.
- **Complex Installation:** Managing numerous connections is difficult.
- **Difficult Maintenance:** Troubleshooting becomes complex as network size increases.
- **Poor Scalability:** Not economical for large networks with many nodes.
- **High Space Requirement:** Bulk wiring requires specialized infrastructure.
- **Complex Routing Logic:** Each node must support routing functionality.

Applications of Mesh Topology

- **Internet Backbone and WANs:** ISPs use partial mesh to connect regional switching centers via high-capacity fiber optics, ensuring high availability and redundancy.

- **Military and Defense Communication Systems:** Used where reliability is mission-critical.
- **Wireless Mesh Networks (WMN):** Utilized in city-wide Wi-Fi and IoT protocols like Zigbee, where wireless nodes relay data for each other to extend range without physical wires.
- **Telecommunication Core Networks:** Used to connect Mobile Switching Centers (MSCs) and tandem exchanges within carrier networks.
- **Emergency Communication Networks:** Maintains connectivity during natural disasters or failures.

3.8 Ring Topology

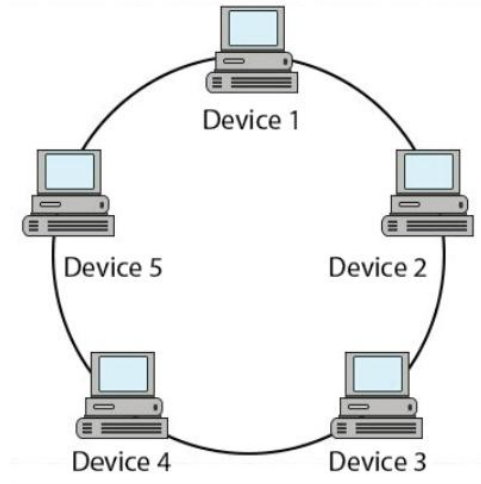
Ring Topology is a network topology in which all devices are connected in a closed-loop (circular) manner. Each device is connected to exactly two neighboring devices, forming a continuous, closed-loop circular data path. Data travels from one node to the next in a predefined direction, making communication orderly and predictable. Ring topology was widely used in early Local Area Networks (LANs) such as Token Ring and FDDI, and is still relevant in specific backbone and industrial networks.

Definition

Ring Topology is a network configuration in which each node (device) is connected to two adjacent nodes, forming a single continuous circular path (a ring) for data transmission. The last device is connected back to the first, completing the ring. Data typically flows in one direction (unidirectional), with each intermediate node acting as a repeater to regenerate and forward the signal.

Structure of Ring Topology

- Devices are connected one after another in a circular (ring) fashion.
- Each node has two point-to-point links, one to each immediate neighbor.
- There is no central controlling device such as a hub or switch.
- The network forms a closed-loop structure with no beginning or end.



Working of Ring Topology

Data Transmission

- Data is transmitted in the form of frames.
- Frames travel from one node to the next around the ring.
- Each node:
 - Receives the frame
 - Checks the destination address
 - Regenerates and forwards it to the next node if it is not the destination
- Transmission continues until the frame reaches the intended destination.

Direction of Data Flow

- Traditionally, data flows in one fixed direction (clockwise or anticlockwise).
- This unidirectional flow avoids data collisions.
- Such a ring is called a simplex or unidirectional ring.

Token Passing Mechanism

- **Token Circulation:** A special control frame called a "token" circulates around the idle ring.
- **Capture and Transmit:** A node wishing to send data must wait for and capture the free token. It then converts the token into a data frame, attaches the message and destination address, and sends it onto the ring.

- **Data Relay:** Each subsequent node receives the frame, checks the address, regenerates the signal, and forwards it to the next node if it is not the intended destination.
- **Destination Receipt:** The destination node recognizes its address, copies the data, marks the frame as received, and puts it back on the ring.
- **Acknowledgment and Token Release:** The frame travels back to the original sender, which removes it from the ring, confirms delivery, and then releases a new token for the next node.

Key Characteristics

- **Closed Loop Structure:** Forms a circular data path with no endpoints.
- **Point-to-Point Links:** Each device has a dedicated connection only to its two immediate neighbors.
- **Deterministic Access:** Uses a controlled access method, typically token passing, which grants each node a turn to transmit, preventing data collisions.
- **Active Topology:** Each node actively participates by receiving, regenerating, and retransmitting the signal, acting as a repeater.
- **Predictable Performance:** The fixed path and token-based control provide consistent, predictable network delays.
- **Equal Access:** All nodes get an equal opportunity to transmit data.
- **Unidirectional/Bidirectional Flow:** Simple rings are unidirectional. Advanced versions like Dual Ring (e.g., FDDI, SONET) use a second, counter-rotating ring for redundancy and self-healing.

Advantages of Ring Topology

- **Collision-Free Communication:** Token passing ensures only one node transmits at a time, eliminating data collisions and providing efficient performance under heavy load.
- **Orderly Data Transmission:** Fixed direction of data flow improves organization.
- **Equal Access for All Nodes:** No node can dominate the network.
- **Better Performance than Bus Topology:** Performs well under heavy network load.
- **Signal Regeneration:** Each node boosts the signal, allowing longer network distances.

- ***Fairly Simple Installation:*** Requires less cabling than a full mesh topology and has no central device to configure.
- ***Suitable for Longer Distances:*** Signal regeneration at each node allows the network to cover greater distances without degradation.

Disadvantages of Ring Topology

- ***Single Point of Failure:*** Failure of one node or cable can disrupt the entire network.
- ***Difficult Troubleshooting:*** Identifying the exact fault location is time-consuming.
- ***Network Disruption During Changes:*** Adding or removing a node requires breaking the ring.
- ***Increased Latency:*** Data must pass through multiple intermediate nodes.
- ***Limited Scalability:*** As the number of nodes increases, token circulation time increases.
- ***Slower than Modern Switched Networks:*** The sequential, store-and-forward nature introduces more latency compared to direct switching in a star topology.

Applications of Ring Topology

- ***Token Ring Networks:*** Older LAN implementations developed by IBM during the 1980s-1990s.
- ***FDDI (Fiber Distributed Data Interface):*** Used a dual-ring topology to create fault-tolerant, high-speed backbones for campuses and metropolitan areas.
- ***Metropolitan Area Networks (MANs):*** Used for city-wide optical fiber communication.
- ***Telecommunication Backbones:*** SONET/SDH rings form the self-healing backbone of modern telephone and carrier networks, providing automatic recovery from fiber cuts.
- ***Industrial Automation Systems:*** Used where predictable and synchronized communication is required.

3.9 Hybrid Topology

Hybrid Topology is a practical and flexible network design that combines two or more different basic topologies (like Star, Ring, Bus, or Mesh) to form a single larger network. It is commonly adopted in large-scale, modern networks where a single topology cannot meet requirements related to performance, reliability, scalability, and cost. It is the most realistic approach for modern, large-scale networks, as it allows designers to leverage the specific strengths of each topology while minimizing their weaknesses. This approach is standard in enterprise, campus, and telecommunication networks where diverse requirements for cost, performance, and reliability must be met simultaneously. In modern telecommunications, the Internet serves as the ultimate example of a hybrid topology, connecting millions of smaller LANs, MANs, and WANs via the TCP/IP protocol suite.

Definition

A Hybrid Topology is a network topology formed by integrating two or more different basic topologies such as Star, Ring, Bus, or Mesh into a single, organized communication system. Each segment retains the characteristics of its original topology. The hybrid design aims to utilize the advantages of different topologies while overcoming their individual limitations.

Structure of Hybrid Topology

The structure of a hybrid network is hierarchical and segmented, allowing for customized deployment based on organizational needs such as physical layout, and traffic demands.

- **Segmented Design:** The network is divided into smaller, manageable sections. Each segment functions as a standalone topology (e.g., a Star segment for an office floor).
- **Interconnecting Devices:** These segments are linked using specialized networking hardware, including:
 - **Routers:** For path selection and connecting different network layers.
 - **Switches/Bridges:** For connecting segments at the data link layer.
 - **Gateways:** To handle protocol translation between different network protocols.
- **High-Speed Backbone:** Usually, a high-performance topology (like a Fiber-Optic Mesh or Ring) acts as the central "spine," carrying heavy traffic between localized hubs.

Working Principle of Hybrid Topology

Hybrid topology functions as a ‘network of networks.’

- **Intra-Segment Communication:** Within each segment, data transmission follows the rules of its native topology. For example, in a Star segment, all traffic passes through the central switch.
- **Inter-Segment Communication:** When a device in one segment needs to communicate with a device in another (e.g., from a Star segment to a Ring segment), the data is handed off to a gateway device—such as a router, bridge, or backbone switch—that interconnects the segments.
- **Backbone Transport:** The interconnected segments often rely on a high-capacity backbone and use a hierarchical design. Local networks (Stars) operate at the edge and a robust backbone (Mesh or Ring) operates at the core.

Key Characteristics of Hybrid Topology

- **Integration of Multiple Topologies:** Combines different topologies such as Star–Ring, Star–Bus, or Mesh–Star.
- **High Scalability:** New segments with different topologies can be added without a complete redesign of the existing network.
- **Structural Flexibility:** Can be customized based on users, traffic load, and building layout.
- **Hierarchical Structure:** Typically features a robust, high-speed backbone (often a Ring or partial Mesh) that connects various departmental segments (often Stars).
- **Segment-Based Fault Isolation:** A failure is usually contained within its segment, preventing a total network collapse.
- **Massive Traffic Handling:** Traffic load is distributed across multiple sub-networks.
- **High Complexity:** Requires advanced planning and skilled network management.
- **Heterogeneous Media Support:** Allows for a mixture of transmission media, such as Unshielded Twisted Pair (UTP) for workstations and Optical Fiber for long-distance backbone links.

Advantages of Hybrid Topology

- **Enhanced Reliability and Fault Tolerance:** A failure in one segment (e.g., a broken cable in a Star) does not bring down the entire network. Critical areas can be designed with redundant topologies like Mesh.
- **High Scalability:** The network can grow organically by adding new segments, making it ideal for expanding organizations.
- **Optimal Performance:** Allows the use of high-performance topologies for critical areas (e.g., Mesh for data centers) and simpler, cost-effective topologies for user access (e.g., Star for desktops), optimizing overall network performance.
- **Customized Design:** Can be tailored to the physical layout, budget, and technical requirements of an organization.
- **Flexibility in Design:** Supports mixed transmission media such as twisted pair and optical fiber.
- **Easier Troubleshooting (Segmented):** Problems can often be isolated to a specific topological segment, simplifying diagnosis.

Disadvantages / Limitations of Hybrid Topology

- **High Initial Cost:** Involves significant investment in extensive cabling and multiple networking devices (switches, routers, bridges).
- **High Complexity:** Design, installation, and configuration are far more complex than for any single basic topology. Requires expert planning.
- **Difficult Maintenance and Management:** Network administrators need expertise in all the incorporated topologies and their interactions. Troubleshooting can be challenging.
- **Design Intensive:** Requires careful initial planning to avoid bottlenecks and ensure efficient communication between different segments.

Applications of Hybrid Topology

- **Large Organizations and Enterprises:** Critical infrastructure like data centers and core routers may be connected in a Mesh, while employee workstations on each floor are connected in a Star topology, all forming one hybrid network.
- **University and Campus Networks:** Different buildings (library, labs, offices) use topologies suited to their function, interconnected by a high-speed fiber-optic backbone (often a Ring or Mesh).

- **Internet Service Providers (ISPs):** The core infrastructure uses a partial Mesh for redundancy, which connects to regional access networks (typically Star topologies) that serve end-users.
- **Telecommunication Networks:** Combination of LANs, MANs, and WANs.
- **Banking and Financial Networks:** Mesh topology for critical data centers connected to Star networks at branches.
- **The Internet:** The ultimate example of a hybrid topology, consisting of millions of interconnected networks (LANs, MANs, WANs) using various topologies, all unified by the TCP/IP protocol suite.

3.10 Overview of Switching and Routing

In telecommunication networks, data must be transferred efficiently from a source to a destination across a large number of interconnected devices. This is achieved using two fundamental mechanisms: Switching and Routing. They perform distinct but complementary functions: Switching focuses on creating a path or forwarding data *within* a network segment, while Routing determines the best path for data to travel *between* different networks. Together, these mechanisms eliminate the need for impractical point-to-point physical connections between every device, allowing for a scalable, global communication infrastructure. Understanding their differences is fundamental to grasping how both traditional systems like the PSTN and modern systems like the Internet operate.

3.11 Switching: Connection Establishment

Switching is the process of connecting communication channels to establish a temporary or permanent path between a sender and a receiver for information exchange. It eliminates the need for a dedicated direct link between every pair of devices, allowing efficient sharing of network resources. In simple terms, switching decides how a connection is established between communicating devices.

Need for Switching

- Direct point-to-point connections between all users are impractical and uneconomical.
- Switching allows multiple users to share network resources.
- Improves utilization of transmission links.

- Makes large-scale networks such as telephone systems and data networks possible.

Types of Switching

There are three primary methodologies used to switch data across a network:

1. Circuit Switching

- **Process:** A dedicated physical communication path is established between the sender and receiver **before** data transfer begins, maintained for the entire session, and released afterwards.
- **Phases:** This involves three distinct phases: Call Setup → Data Transfer → Call Teardown.
- **Characteristics:** Provides guaranteed bandwidth throughout the session and constant delay (low jitter). Once established, there is no congestion on the established path.
- **Efficiency:** Low; the circuit remains occupied during idle periods (e.g., silence in a voice call).
- **Advantages:** Constant delay, ideal for real-time voice communication, reliable and predictable performance.
- **Limitations:** Inefficient for bursty data (line remains idle during silence), Connection setup time is required.
- **Primary Application:** Public Switched Telephone Network (PSTN) for voice calls.

2. Message Switching (Historical)

- **Process:** The entire message is treated as a single data unit and transferred using a store-and-forward mechanism. Each intermediate node receives, stores, and then forwards the complete message to the next node.
- **Characteristics:** No dedicated path is established. Causes large delays due to storage at each hop. Not suitable for real-time communication. Used in old telegraph networks.
- **Primary Application:** Telegraph networks (rarely used in modern real-time systems).

3. Packet Switching

- **Process:** Data is divided into small units called packets. Each packet contains a header with the destination address and is

transmitted independently. Packets may take different paths through the network and arrive out of order.

- **Sub-types:**

- **Datagram Approach (Connectionless):** Packets are routed independently; they may take different paths and arrive out of order (e.g., IP).
- **Virtual Circuit Approach (Connection-oriented):** A logical path is pre-defined; all packets follow the same route, ensuring sequential delivery (e.g., MPLS).
- **Characteristics:** Highly efficient use of network links (statistical multiplexing). Robust, with dynamic alternate routing if a link fails. Can suffer from variable delay (jitter) and packet loss. Requires protocols for sequencing, flow control, and error control (e.g., TCP).
- **Primary Application:** The Internet and all modern data networks.

3.12 Routing: Path Determination

Routing is the process of selecting the best path for data packets to travel from the source to the destination across a network. It is the core function that directs traffic on the Internet.

The Router and Routing Table

The Router is a specialized device that connects multiple networks. Its operation relies on two key functions:

- **Path Determination:** Uses routing protocols to build and maintain a routing table—a map of the network. Routers use routing tables and algorithms to decide the optimal path.
- **Packet Forwarding:** Examines a packet's destination IP address, consults the routing table, and sends the packet out the appropriate interface toward the next hop.

The routing table is a database in the router's memory containing the destination network addresses, next-hop addresses (the address of the next router in the path) and interfaces (the physical ports used to exit the router).

Types of Routing

- **Static Routing:** The network administrator manually configures routes. Simple and secure but not scalable.

- **Dynamic Routing:** Routers automatically share information via protocols to adapt to congestion or link failures.
 - **Interior Gateway Protocols (IGP):** Used within a single Autonomous System (AS). **Examples: OSPF, RIP, EIGRP.**
 - **Exterior Gateway Protocols (EGP):** Used to route data between different organizations/ASes. **Example: BGP (Border Gateway Protocol).**

The Routing Process

1. A router receives a packet.
2. It checks the destination IP address against its routing table.
3. It finds the longest matching network entry.
4. It forwards the packet to the next-hop router indicated in the table via the appropriate interface.
5. This process repeats at each router (hop) until the packet reaches its final destination network.

Comparison: Switching Vs Routing

Feature	Switching	Routing
Primary Function	Establishes a dedicated connection/path between devices within the same network.	Selects the optimal path and forwards packets between different networks.
Resource Usage	Dedicated (reserved) resources throughout the session.	Shared (statistical) resources.
Data Unit	Continuous stream or Frames.	Packets.
Addressing	Physical/Hardware/Phone Number.	Logical IP Address (e.g., 192.168.1.1).
Path Determination	Determined once at call setup using hardware.	Determined per-packet, at each router (hop-by-hop) using software.
Hardware	Switch / Telephone Exchange.	Router.
Speed	Generally faster as it uses hardware-based switching (frame forwarding).	Slower than switching as it involves software-based IP lookup and complex decision-making.
Error Handling	Basic error checking (CRC). No end-to-end error recovery.	Relies on higher-layer protocols (like TCP) for end-to-end error control and recovery.
Typical Network	LAN, PSTN.	WAN, Internet.
Example	Connecting multiple computers in an office using an Ethernet switch.	Connecting your home network to the Internet via an ISP's router.

Comparison: Circuit Switching vs. Packet Switching

Parameter	Circuit Switching	Packet Switching
Basic concept	A dedicated physical path is established between the sender and receiver before transmission.	Data is divided into small units called packets, which are transmitted independently.
Connection setup	Required before communication starts	Not required
Path used	The path is fixed for the entire duration of the communication.	The path is dynamic; each packet can potentially take a different route.
Resource utilization	Bandwidth and network resources are reserved in advance (Pre-allocation).	No prior reservation; resources are allocated on-demand (Dynamic allocation).
Efficiency	Low: Resources are wasted during silent periods (idle time) as the line is blocked.	High: Multiple users share the same channel, leading to better resource utilization.
Data transmission	Data arrives in the same sequence it was sent.	Packets may arrive out of order and must be reassembled at the destination.
Delay	Initial setup delay, but constant transmission delay	No setup delay, but variable delay due to congestion
Reliability	Very reliable once the circuit is established	Reliability depends on network conditions
Suitability	Best for real-time voice communication	Best for data and internet communication
Example	PSTN (Traditional telephone network)	Internet

Comparison of Network Types (PSTN, Cellular, Satellite, and the Internet)

Parameter	PSTN	Cellular Network	Satellite Network	Internet Network
Basic purpose	Voice communication	Mobile voice & data	Long-distance communication	Global data communication
Transmission medium	Wired (copper/fiber)	Wireless (radio waves)	Microwave via satellite	Wired + wireless
Coverage area	Local / national	Wide area (cells)	Very wide / global	Global
Mobility	Not supported	Supported	Limited	Limited
Switching technique	Circuit switching	Circuit + Packet	Packet switching	Packet switching
Infrastructure	Fixed lines, exchanges	BTS, MSC	Satellite, earth station	Routers, servers
Delay	Low	Low-moderate	High	Variable
Cost	High maintenance	Moderate	Very high	Low-moderate
Common applications	Landline calls	Mobile phones, data	TV broadcast, GPS	Email, web, cloud

Comparison: Network Topologies (Star Topology, Ring Topology, Mesh Topology and Hybrid Topology)

Parameter	Star Topology	Ring Topology	Mesh Topology	Hybrid Topology
Basic structure	All nodes connected to a central device	Nodes connected in a closed loop	Each node connected to multiple or all nodes	Combination of two or more topologies
Central device	Required (hub/switch)	Not required	Not required	Depends on constituent topologies
Number of paths	Single path via central device	Single fixed path	Multiple paths	Multiple (depends on design)
Fault tolerance	Moderate	Low	Very high	High
Effect of single node failure	Only that node affected	Entire network may fail	Network continues to work	Limited to affected segment
Installation complexity	Simple	Moderate	Very complex	Complex
Cost	Moderate	Low to moderate	Very high	High
Scalability	Easy	Difficult	Difficult	Very good
Performance	Good	Predictable	Excellent	Good to excellent
Typical applications	LANs, offices	Token Ring, MANs	Backbone, military	Large enterprises, campuses

Review Questions

PSTN (Public Switched Telephone Network)

1. Define PSTN and explain its hierarchical architecture with a neat block diagram.
2. Explain the working principle of PSTN. Describe call setup, call holding, and call release phases.
3. Explain the advantages, disadvantages, and applications of PSTN.

Cellular Networks

4. Draw and explain the architecture of a cellular network. Explain the functions of MS, BTS, BSC, MSC, HLR, VLR, AuC, and GMSC.
5. Explain frequency reuse and handoff in cellular networks. Describe the need for handover. Explain hard handoff and soft handoff.
6. Explain the working principle of a cellular network. Describe the complete call flow from call initiation to call termination.

Satellite Networks

7. Explain satellite networks with a neat block diagram. Describe their definition, necessity, and key characteristics.
8. Describe the components of a satellite network. Explain space segment, ground segment, and control segment.
9. Explain the working principle of satellite communication. Describe uplink, transponder operation, and downlink.
10. Explain the types of satellite orbits with applications.
11. Explain the advantages, disadvantages, and applications of satellite networks.

Internet Networks

12. Explain Internet networks with reference to their definition, characteristics, and decentralized architecture.
13. Explain the architecture and components of Internet networks and describe the working principle of Internet communication.
14. Explain the advantages, disadvantages, and applications of Internet networks.
15. Compare Network Types (PSTN, Cellular, Satellite, and the Internet).

Network Topologies

15. Explain network topology. Distinguish between physical topology and logical topology.
16. Explain star topology with a neat diagram. Describe its structure, working, advantages, disadvantages, and applications.
17. Explain mesh topology in detail. Describe its types, working, advantages, disadvantages, and applications.
18. Explain ring topology with reference to the token passing mechanism. Describe its working, advantages, disadvantages, and applications.
19. Explain hybrid topology. Describe its structure, working principle, advantages, disadvantages, and applications.
20. Compare star, ring, mesh and hybrid topologies.

Switching

21. Explain switching in telecommunication networks. Describe the need for switching and its role in large-scale networks.
22. Explain circuit switching in detail. Describe its working, characteristics, advantages, limitations, and applications.
23. Explain packet switching in detail. Describe its working, types, characteristics, advantages, and applications.
24. Compare circuit switching and packet switching.

Routing

25. Explain routing in computer networks. Describe the role of routers and routing tables.
26. Explain the routing process in detail. Describe static routing and dynamic routing.
27. Compare switching and routing.